

## Reservoir Sandstone Stratigraphic Correlation

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*This lab activity allows students to construct stratigraphic models and use them to predict lithologic differences across the Appalachian basin. Then students are asked to evaluate areas for a prospective CO<sub>2</sub> subsurface storage project*

### Introduction (with geoscience context)

One of the strategies for mitigating CO<sub>2</sub> emissions involves injecting captured CO<sub>2</sub> underground into porous rock reservoirs. You will consider rock core data and geologic models to assess potential locations of Upper Devonian Hampshire Fm. sandstone that could serve as underground storage reservoirs for CO<sub>2</sub>.

### Learning Goals

- Develop stratigraphic column construction skills
- Create core models that can be used to predict stratigraphic variability that can influence storage reservoir rock
- Infer stratigraphic variability from basin context

### Pre-activity assignments/reading (if applicable)

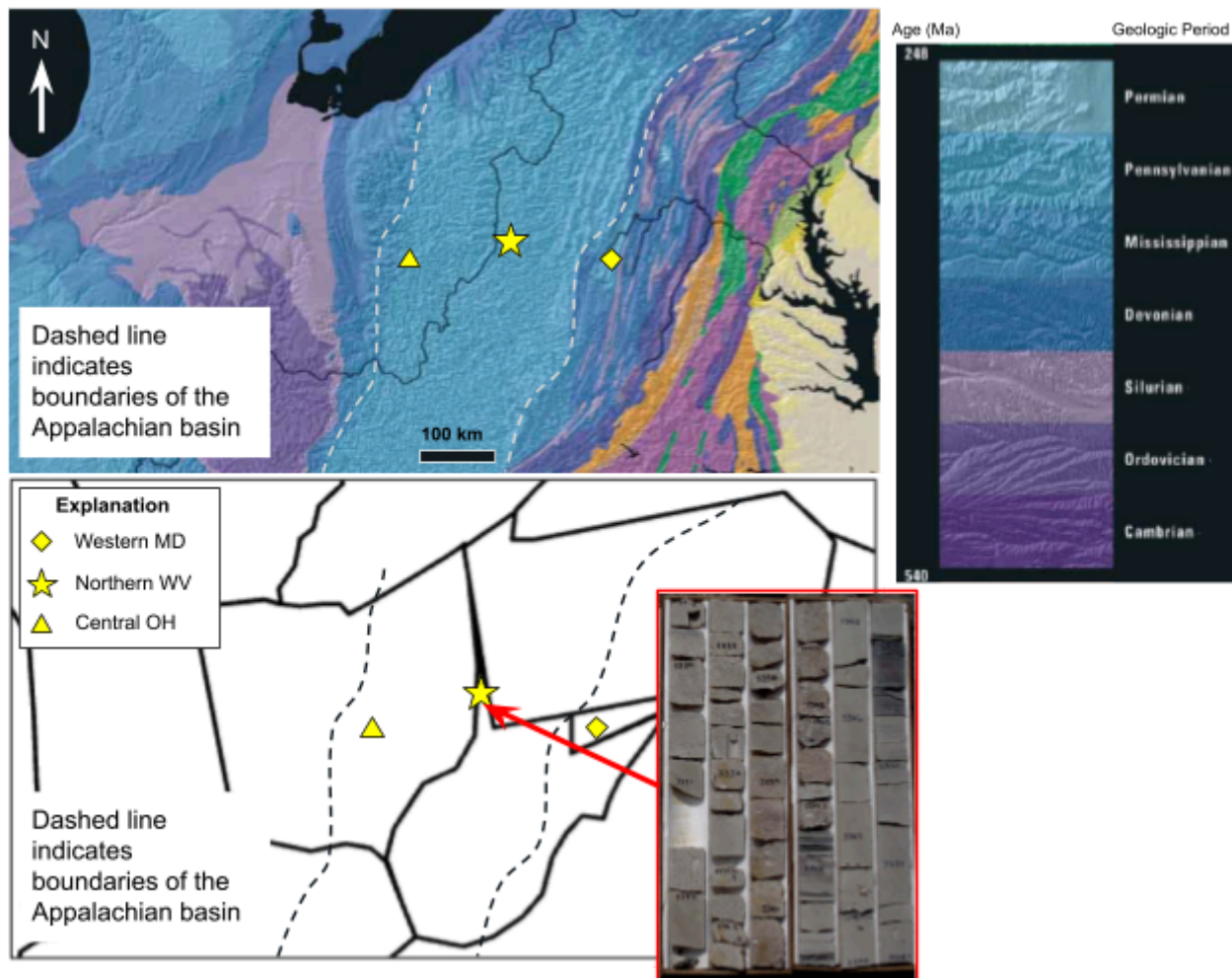
- [Bentley et al. Historical Geology free online textbook 4.2.1 Formations and Walther's Principle](#)
- [Energy Education Encyclopedia: Oil and Gas Traps](#)

### Using the Visible Geology Tools

Visit the [Visible Geology app](#) webpage to create diagrams for this assignment. [Refer to this Youtube Tutorial for help](#).

## Activity (description and steps)

### Part 1: Core Column Construction



**Instructions:** A core was recently obtained at the location of the yellow star in the map where a well was drilled to >6000 ft and recovered about 30 ft. of Upper Devonian Hamshire Formation. Description of the Upper Devonian core was performed by the project geologist in the early 2000's and this description was archived by the West Virginia Geological and Economic Survey.

**1) Use the archived core description in the Table provided in the following pages to construct a stratigraphic column for the Northern WV core.** Note the core description starts with the oldest/bottommost sedimentary bed of the core (the deepest rock that was recovered); this means the subsequent beds in the table are found upward from that bottom bed, so using the Principle of Superposition the beds are listed in Table 1 in progressively younger order.

The beds are divided into 3 Stratigraphic Units based on the interpreted depositional environment, listed from oldest to youngest:

- **Strat Unit 1:** from 0 - 6.7 feet formed in an offshore environment
- **Strat Unit 2:** from 6.7 - 18.05 feet formed in a nearshore environment
- **Strat Unit 3:** from 18.05 - 30.05 feet formed in a delta plain channel environment



# Visible Geology

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**Table 1.** Sedimentologic Description of the Hampshire Fm. Core from Northern West Virginia

| Bed #                 | Bed Thickness (ft) | Sum Thickness (ft) | Lithology                                | Sedimentary Structure                                                                                                                     | Environ. of deposition |                     |
|-----------------------|--------------------|--------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------|
| <b>BOTTOM OF CORE</b> |                    |                    |                                          |                                                                                                                                           |                        |                     |
| 1                     | 0.1 ft             | 0.1                | Dark gray shale                          | Fissile lamination                                                                                                                        | Offshore               | <b>STRAT UNIT 1</b> |
| 2                     | 1.5 ft             | 1.6                | Very fine grained sandstone              | Hummocky cross stratification                                                                                                             | Offshore               |                     |
| 3                     | 1.4 ft             | 3                  | Fine-grained sandstone                   | Hummocky cross stratification & horizontal stratification                                                                                 | Offshore               |                     |
| 4                     | 0.6 ft             | 3.6                | Fine-grained sandstone                   | Hummocky cross stratification & horizontal stratification                                                                                 | Offshore               |                     |
| 5                     | 0.7 ft             | 4.3                | Fine-grained sandstone                   | Hummocky cross stratification, mud rip-up clasts                                                                                          | Offshore               |                     |
| 6                     | 2.25 ft            | 6.55               | Fine-grained sandstone                   | horizontal lamination                                                                                                                     | Offshore               |                     |
| 7                     | 0.15 ft            | 6.7                | Dark gray shale                          | Fissile lamination                                                                                                                        | Offshore               |                     |
| 8                     | 1.1 ft             | 7.8                | Fine-grained sandstone                   | horizontal stratification                                                                                                                 | Nearshore              | <b>STRAT UNIT 2</b> |
| 9                     | 1.0 ft             | 8.8                | Fine-grained sandstone                   | Hummocky cross stratification; rare rounded quartz pebbles ( $\approx 0.2$ inch diameter) & mud rip-up clasts ( $\leq 0.1$ inch diameter) | Nearshore              |                     |
| 10                    | 0.2 ft             | 9                  | Pebble conglomerate, average size 0.4 cm | mud rip-up clasts ( $\leq 0.1$ inch diameter); gravel in matrix of fine-grained sand                                                      | Nearshore              |                     |
| 11                    | 0.4 ft             | 9.4                | MISSING CORE                             |                                                                                                                                           | Nearshore              |                     |
| 12                    | 0.4 ft             | 9.8                | Medium-grained sandstone                 | Cross-lamination, mud rip-up clasts ( $\leq 0.1$ inch diameter)                                                                           | Nearshore              |                     |
| 13                    | 0.15 ft            | 9.95               | Dark gray shale                          | Fissile lamination                                                                                                                        | Nearshore              |                     |
| 14                    | 2.4 ft             | 12.35              | Fine-grained sandstone                   | Horizontal stratification                                                                                                                 | Nearshore              |                     |
| 15                    | 0.15 ft            | 12.5               | Dark gray shale                          | Fissile lamination                                                                                                                        | Nearshore              |                     |
| 16                    | 0.55 ft            | 13.05              | Fine-grained sandstone                   | Hummocky cross stratification                                                                                                             | Nearshore              |                     |
| 17                    | 1.0 ft             | 14.05              | Dark gray shale                          | Fissile lamination                                                                                                                        | Nearshore              |                     |
| 18                    | 2.6 ft             | 16.65              | Fine-grained sandstone                   | Horizontal stratification, few scattered quartz pebbles ( $\approx 0.2$ inch diameter)                                                    | Nearshore              |                     |
| 19                    | 0.8 ft             | 17.45              | Fine-grained sandstone                   | Horizontal stratification, few scattered quartz pebbles ( $\approx 0.2$ inch diameter)                                                    | Nearshore              |                     |
| 20                    | 0.6 ft             | 18.05              | Fine-grained sandstone                   | Cross-stratification; thin (0.5 inch                                                                                                      | Nearshore              |                     |



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|    |         |       |                                          |                                                                                                                                       |                     |              |
|----|---------|-------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------|
|    |         |       |                                          | thick) horizon of quartz pebbles ( $\approx 0.2$ inch diameter) and mud rip-up clasts ( $\leq 0.1$ inch diameter)                     |                     |              |
| 21 | 2.7 ft  | 20.75 | Fine-grained sandstone                   | Horizontal stratification; rare rounded quartz pebbles ( $\approx 0.2$ inch diameter) & mud rip-up clasts ( $\leq 0.1$ inch diameter) | Delta plain channel | STRAT UNIT 3 |
| 22 | 0.3 ft  | 21.05 | Pebble conglomerate, average size 0.4 cm | Horizontal stratification; mud rip-up clasts ( $\leq 0.1$ inch diameter); gravel in matrix of medium- to coarse-grained sand          | Delta plain channel |              |
| 23 | 1.7 ft  | 22.75 | Medium-grained sandstone                 | Horizontal stratification; rare rounded quartz pebbles ( $\approx 0.2$ inch diameter)                                                 | Delta plain channel |              |
| 24 | 0.2 ft  | 22.95 | Pebble conglomerate, average size 0.8 cm | gravel in matrix of very coarse-grained sand                                                                                          | Delta plain channel |              |
| 25 | 0.2 ft  | 23.15 | Coarse-grained sandstone                 | Horizontal stratification; rare mud rip-up clasts ( $\leq 0.1$ inch diameter)                                                         | Delta plain channel |              |
| 26 | 0.3 ft  | 23.45 | Medium-grained sandstone                 | Horizontal stratification; mud rip-up clasts throughout ( $\leq 0.1$ inch diameter)                                                   | Delta plain channel |              |
| 27 | 1.6 ft  | 25.05 | Very coarse-grained sandstone            | Cross-stratification; rare scattered quartz pebbles ( $\approx 0.2$ inch diameter) throughout bed                                     | Delta plain channel |              |
| 28 | 0.8 ft  | 25.85 | Medium-grained sandstone                 | Horizontal stratification                                                                                                             | Delta plain channel |              |
| 29 | 0.7 ft  | 26.55 | Coarse-grained sandstone                 | Cross-stratification; rare scattered quartz pebbles ( $\approx 0.2$ inch diameter) throughout bed                                     | Delta plain channel |              |
| 30 | 0.2 ft  | 26.75 | Medium-grained sandstone                 | Cross-stratification; rare quartz pebbles ( $\approx 0.2$ inch diameter)                                                              | Delta plain channel |              |
| 31 | 1.85 ft | 28.6  | Coarse-grained sandstone                 | Cross-stratification; rare scattered quartz pebbles ( $\approx 0.2$ inch diameter) throughout bed                                     | Delta plain channel |              |
| 32 | 0.25 ft | 28.85 | Medium-grained sandstone                 | Cross-stratification                                                                                                                  | Delta plain channel |              |
| 33 | 0.3 ft  | 29.15 | Coarse-grained sandstone                 | No evident structures                                                                                                                 | Delta plain channel |              |
| 34 | 0.7 ft  | 29.85 | Medium-grained sandstone                 | Trough cross-stratification                                                                                                           | Delta plain channel |              |
| 35 | 0.2 ft  | 30.05 | Coarse-grained sandstone                 | Cross-stratification                                                                                                                  | Delta plain channel |              |

**TOP OF CORE**



# Visible Geology

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| Scale (ft) |                                   |                              |        | Sedimentary features | Depositional Environment |
|------------|-----------------------------------|------------------------------|--------|----------------------|--------------------------|
|            | MUD                               | SAND                         | GRAVEL |                      |                          |
|            | clay<br>silt<br>vf<br>f<br>m<br>c | gran<br>pebb<br>cobb<br>boul |        |                      |                          |
| 0          |                                   |                              |        |                      |                          |
| 2          |                                   |                              |        |                      |                          |
| 4          |                                   |                              |        |                      |                          |
| 6          |                                   |                              |        |                      |                          |
| 8          |                                   |                              |        |                      |                          |
| 10         |                                   |                              |        |                      |                          |
| 12         |                                   |                              |        |                      |                          |
| 14         |                                   |                              |        |                      |                          |
| 16         |                                   |                              |        |                      |                          |
| 18         |                                   |                              |        |                      |                          |
| 20         |                                   |                              |        |                      |                          |
| 22         |                                   |                              |        |                      |                          |
| 24         |                                   |                              |        |                      |                          |
| 26         |                                   |                              |        |                      |                          |
| 28         |                                   |                              |        |                      |                          |
| 30         |                                   |                              |        |                      |                          |

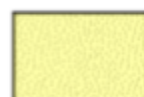
## EXPLANATION OF PATTERNS AND SYMBOLS



Conglomerate



Medium-Coarse Sandstone



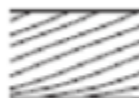
Fine Sandstone



Mudstone



missing core



Cross-stratification



Hummocky cross-stratification



Trough cross-stratification



Horizontal stratification



Quartz pebble



Mud rip-up





# Visible Geology

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## Part 2: Lateral Stratigraphic Modeling

The Hampshire Formation was deposited as part of a prograding sediment wedge, including a delta system. This delta system advanced or prograded from the proximal areas in eastern PA westward in modern coordinates across West Virginia toward Ohio. To the right the image shows a model of the WV core created with the Geology Explorer App in Visible Geology. This representation of the core shows the 3 units in the proportion of they appear in the northern WV core:

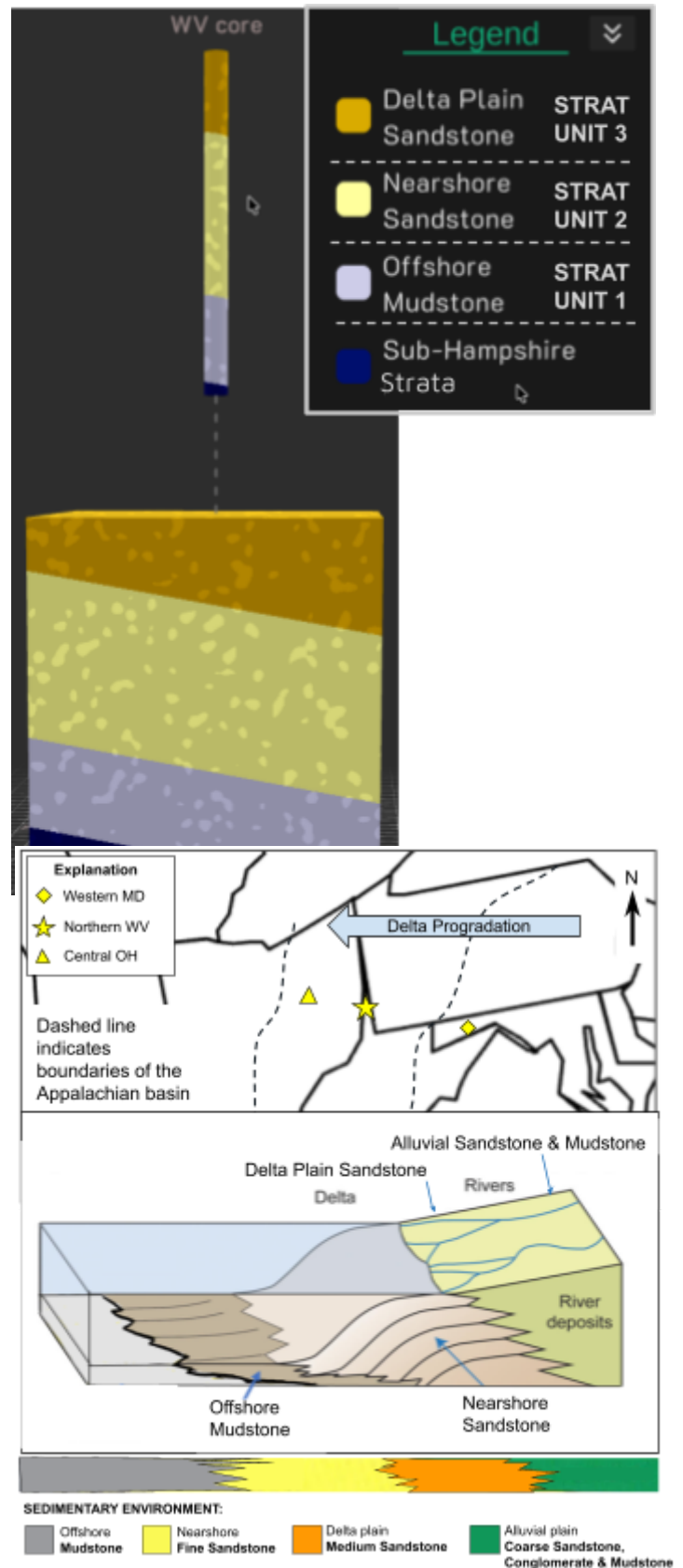
- **Strat Unit 1** is composed of offshore mudstone
- **Strat Unit 2** composed of nearshore fine-grained sandstone
- **Strat Unit 3** composed of delta plain coarse-grained sandstone.

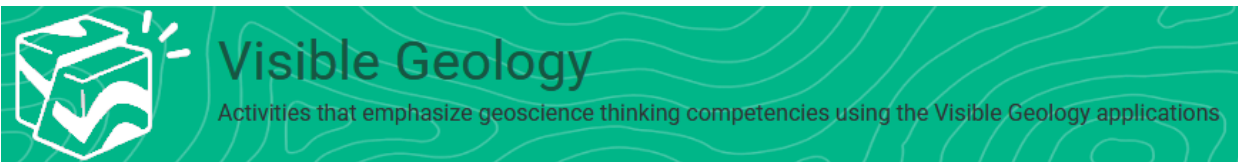
Below the Hampshire units are sub-Hampshire strata.

1) Does the stacking pattern of the lithologies in the core suggest overall shoreline transgression or shoreline regression? Explain your answer:

2) Using the [Geology Explorer app by Visible Geology](#), create a representative core of coeval Hampshire Fm. deposits you would expect to be present in the subsurface of Central OH and Western MD. Your core models should:

- Show the progression of deposits that would result from the **same shoreline movement** as you inferred from the WV Core
- Show units produced by environments of the Hampshire Fm. delta system (offshore, nearshore, delta plain and alluvial plain environments); Note the **proportion** of each unit you predict should differ based on the relative position of the core in the basin and with respect to the direction of delta progradation. Adjust the colors of the layers to reflect the depositional environment associated with each interval; colors should match the colors assigned to environments in the diagram to the left





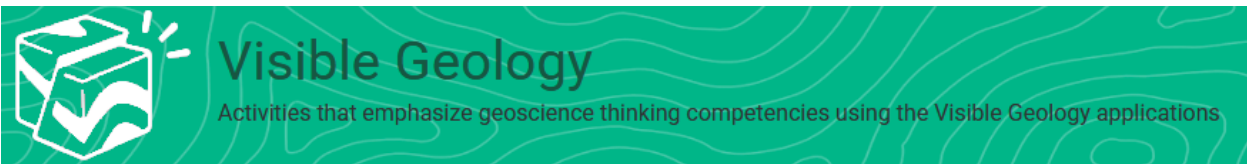
To make your model:

- **Create a new Project in Geology Explorer**
- **Click on Layers to edit/add beds.** To change the color of a bed, click on the pencil icon in the block model. To remove a bed, click on the trash can icon in the block model. To add a bed, click on the **+** icon in the menu in the lower right.
  - The lowermost bed should be the Sub-Hampshire strata; make this layer a dark color.
  - Hampshire stratigraphic units should be created above the Sub-Hampshire strata. You should decide **how many Strat Units you want to show**. There are 4 possible Strat Units:
    - Offshore mudstone (color: gray)
    - Nearshore fine-grained sandstone (color: light yellow)
    - Delta plain channel medium-grained sandstone (color: dark orange)
    - Alluvial plain coarse-grained sandstone, conglomerate and mudstone (color: green)

You should represent the Hampshire Fm. with at least 2 strat units but you do not have to have all 4 present in any one core.

- **Adjust thickness of each bed so that it shows the relative thickness you predict for each Strat Unit at that location.** Click on the  icon in the block model and drag to make the bed thinner or thicker. Depending on the position in the basin and relative position in the delta system, you should predict different thicknesses for each Strat Unit that you have chosen to represent in the Hampshire Fm. Be sure you have considered some logical reason for depicting the relative thickness of the units you show in each model.
- **Make a core.** Once you have your block model appearing the way you want it, click on the Core icon in the menu and add a vertical core. Give the core a title to reflect the location that it represents.

Paste links and screenshots of your core models in the next page.



**Central Ohio Core Model:**

**Link:**

**Screenshot:**

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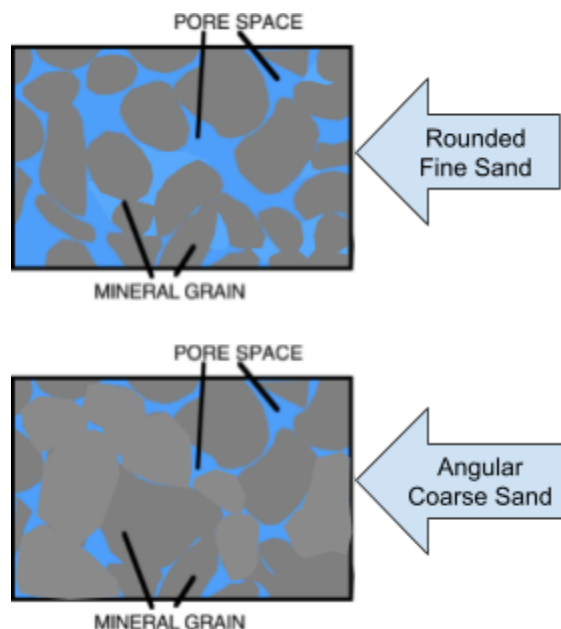
**Western MD Core Model:**

**Link:**

**Screenshot:**



3) Nearshore sandstone offers the best reservoir rock because the well sorted and well rounded grains means there is greater pore space available to store gas (see diagram below); based on this, which area do you think would have the thickest and most abundant Nearshore fine-grained sandstone reservoir rock: Western MD, Northern WV or Central OH? Explain your answer.

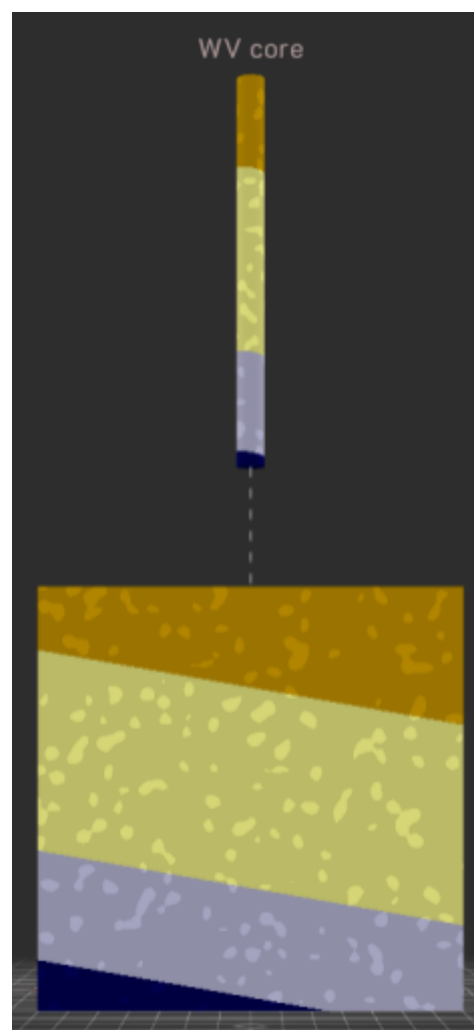


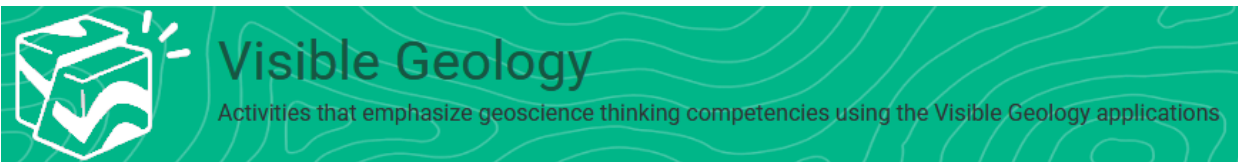
### Part 3: Structural Modeling

To determine if an area may be suitable for subsurface CO<sub>2</sub> sequestration, we will need to evaluate how likely the CO<sub>2</sub> may leak out. Storing gas in a subsurface reservoir requires not only a porous reservoir rock but a “trap”. Read about traps from the [Energy Education Encyclopedia](#). Injected CO<sub>2</sub> will flow through porous rock upward buoyancy due to its lower density than surrounding material. **NOTE:** A thick mudstone overlies the Hampshire Formation in this area, forming an impermeable sealing rock that prevents vertical or upward flow. Therefore, the structural configuration of the Hampshire Fm. will be the primary influence on the direction of flow or migration of the CO<sub>2</sub>.

Consider the WV core as an example. It is located in an area where the strata dip 5° toward the east.

1) [Examine the tilted core model of the Northern WV core in Visible Geology](#); In the Visible Geology app, you can see the strata are shown tilted toward the east with 2X vertical exaggeration to accentuate dip. The tilt of the strata will cause CO<sub>2</sub> injected into this unit to migrate - **what direction would injected CO<sub>2</sub> migrate?**



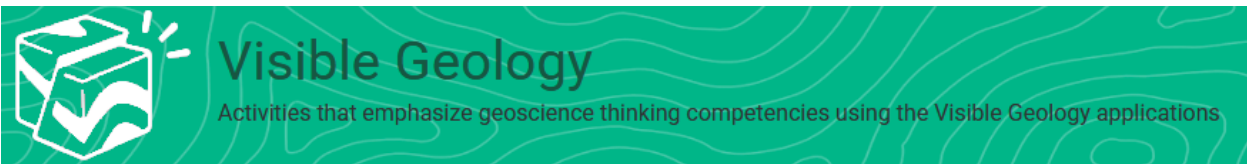


**2) Edit your 2 Visible Geology models for the Western MD and Central OH cores in order to transform the strata into the appropriate structural configuration as described below.**

- **Central OH:** strata dip  $\sim 5^\circ$  toward the east **but show this dip with 2X vertical exaggeration**
- **Western MD:** strata have been deformed by folds with the following attributes as described below; for this location do not use any vertical exaggeration:
  - Fold axis trend and plunge:  $022/10^\circ$
  - Fold amplitude = 16
  - Fold wavelength = 20

**Paste screenshots and links of your modified models for each area on the next page and for each model either describe using words or draw arrows to indicate the direction(s) fluid would flow through the reservoir rock. Remember there is an impermeable sealing rock that overlies the top of your model.** W

**2) Which area provides the best opportunity to trap injected  $\text{CO}_2$  in nearshore sandstone of the Hampshire Fm.? Rank the 3 locations and provide evidence you determined from your core models to support your ranking. Consult the [Energy Education Encyclopedia](#) and explain your reasoning behind your selection choice.**



**Central Ohio:**

**Link:**

**Screenshot:**

**Western MD:**

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**Link:**

**Screenshot:**



# Visible Geology

Activities that emphasize geoscience thinking competencies using the Visible Geology applications

| LEARNING GOAL                                                                                                        | ASSESSMENT CRITERIA                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Develop stratigraphic column construction skills</li></ul>                   | <ul style="list-style-type: none"><li>• Create an accurate sedimentary log following professional conventions</li></ul>                                              |
| <ul style="list-style-type: none"><li>• Infer stratigraphic variability</li></ul>                                    | <ul style="list-style-type: none"><li>• Identify greater proportion of offshore facies toward the west and potential for alluvial plain facies in the east</li></ul> |
| <ul style="list-style-type: none"><li>• Create predictive geologic models for potential storage reservoirs</li></ul> | <ul style="list-style-type: none"><li>• Identify storage opportunities</li></ul>                                                                                     |